CORE SERIES VII

Quantum Seeds and Effective Dynamics

Symmetry, geometric transport, quantum channels, and bounded refinement towers

A quantum seed generates a space through an operator grammar. Packets resolve how symmetry, evolution, geometry, and hidden degrees of freedom change that space.

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How to use this volume

Part I. The finite quantum seed calculus develops the packet language and states the precise scope of the archived formal proof.
Part II. The strengthened development extends this language through seventy-nine theorem, proposition, lemma, and corollary proofs.
Part III. Six supplements give the detailed mathematical constructions behind the formal proofs, followed by three alternative derivations.
Part IV. Forty worked visual pages connect the formulas to finite models and identify the hypotheses needed at each continuum boundary.

Reading this edition

A seed is useful only together with the operators allowed to act on it. Paper VII follows that principle through symmetry packets, quantum circuits, moving projectors, channels, effective dynamics, and refinement. The finite calculus in Part I supplies the common language. The strengthened development in Part II gives broader bounded constructions and limit criteria. Part III retains the detailed proof completions rather than reducing them to a status summary.

The original finite item numbers keep the prefix I. The strengthened development has the prefix II, and the supplementary constructions have the prefix III. All nine mathematical source parts are retained. This edition revises their explanatory prose while preserving the mathematical expressions, hypotheses, proof steps, and statement inventory. The accompanying report records the changes. Overlapping statements and proofs are retained because their hypotheses and uses differ. Their total is not a count of independent discoveries.

The selected archived formal checkpoint covers nineteen scoped manuscript items. It records 323 explicit theorem declarations across twenty-five active Lean proof modules. These are different units. The strengthened Part II is outside that nineteen-item contract. The formal evidence comes from the archived checkpoint of 6 September 2026. The present prose revision does not report a fresh Lean compilation. Exact hypotheses, declarations, inherited compiler records, and reproduction instructions appear in the Lean companion.

Five explicitly named conjectures remain open in the finite core: anomaly-free refinement, uniform Feshbach control, Lorentzian reconstruction, the Einstein infrared limit, and massless spin-two packets. Bounded direct limits and norm limits provide useful criteria. Each physical target still needs the additional structure stated in its conjecture.

Four distinctions to keep in view

A seed's algebraic orbit is different from a unitary state-preparation orbit. Compression is different from an effective resolvent. A normalized channel is different from an unnormalized error family. A finite spectrum is different from a continuum particle interpretation. Keeping these distinctions visible makes the later results easier to use.

Read Part I first for the packet grammar and the finite statements. Turn to Part II for bounded operator transport, spectral and channel constructions, and refinement towers. Use Part III when the geometric bundle construction or a formal specialization needs more detail. The visual atlas follows the same

topics through explicit examples. It illustrates the statements with finite data and does not replace their proofs.

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Part I Corrected finite quantum seed calculus

A route through this section

The finite core starts with seed rank, then resolves symmetry and circuit transfer into packets. Its geometric statements use the actual range bundle, including the Lie-bracket term. Its channel and Feshbach statements keep normalization and invertibility visible. The three examples below introduce those distinctions with exact finite data.

Three exact finite models

A seed remembers multiplicity support

For $M_2(\mathbb{C}) \otimes I_3$, a coefficient matrix with rank one generates an orbit of dimension two. A matrix of rank two generates an orbit of dimension four. Neither single seed spans the full space of dimension six, because a 2×3 matrix cannot have rank three. Two seeds can nevertheless cover all three multiplicity coordinates together. The example illustrates the finite seed rank formula with $d = 2$ and $m = 3$.

Seed rank measures the available multiplicity support

```
C1 = [[1, 0, 0], [0, 0, 0]]

C2 = [[1, 0, 0], [0, 1, 0]]

C3 = [[0, 0, 1], [0, 0, 0]]

C2 and C3 cover all 3 columns.
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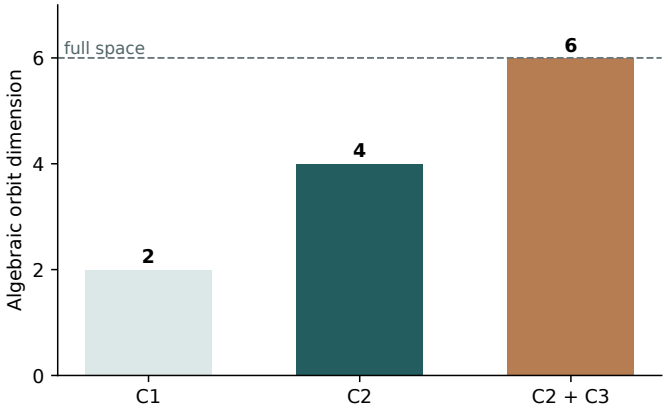


Figure 1. The figure displays exact coefficient matrices for $d = 2, m = 3$. The combined row support of two seeds has rank three. The bar heights record algebraic orbit dimensions and should not be read as probabilities of unitary state preparation.

An eliminated coordinate still contributes

Take $H = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 3 \end{smallmatrix}\right)$ and retain its first coordinate. The eliminated block is invertible and the Schur complement of H is $2 - 1/3 = 5/3$. Its inverse is $3/5$, exactly the retained entry of H^{-1}. Inverting the compressed entry alone gives $1/2$. The difference records the contribution of a path through the eliminated coordinate. Direct compression omits that contribution, while block inversion retains it.

Block inversion includes the discarded coordinate

```
H = [[2, 1], [1, 3]]

Schur complement of H:

2 - 1/3 = 5/3

Retained entry of H inverse:

(5/3) inverse = 3/5
```

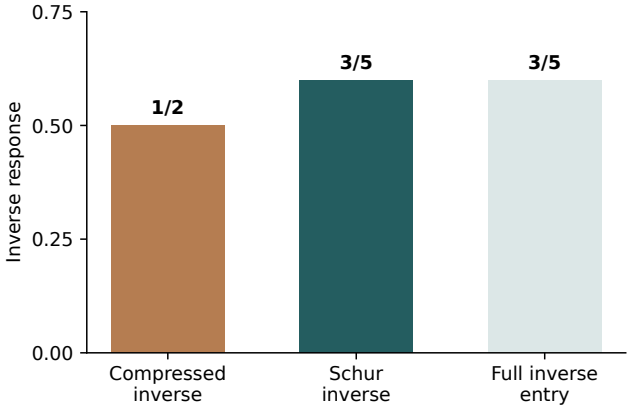


Figure 2. This rational model uses two coordinates to illustrate block inversion. Both the eliminated block and the full operator are invertible, and the plotted values are exact fractions. The example shows why direct compression and resolvent elimination give different results.

Normalization changes the map

Let $P = |0\rangle\langle 0|$ on $\mathbb{C}^2$ and $Z = \mathrm{diag}(1, -1)$. The normalized Kraus family $\sqrt{3/4}\, I$, $\sqrt{1/4}\, Z$ satisfies $\sum_j K_j^* K_j = I$ and maps P to P. The unnormalized family $\sqrt{3}\, I$, Z satisfies $\sum_j E_j^* E_j = 4I$ and maps P to $4P$. The two families have the same relative weights, but their total normalization differs. Both outputs are proportional to the same projector, but their traces differ.

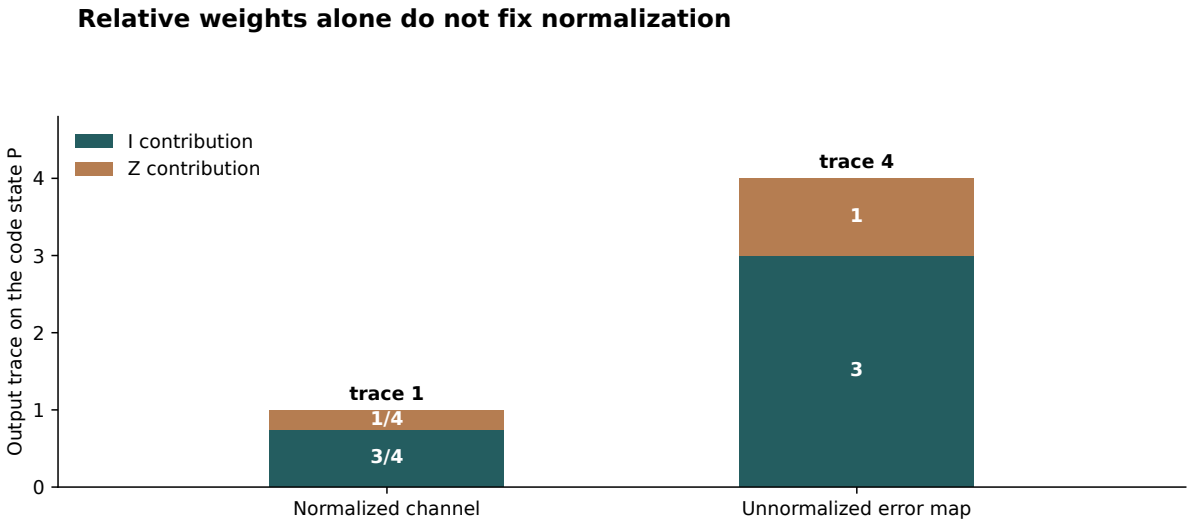


Figure 3. The figure shows exact weights for one supported code state. The normalized map preserves trace one, while the unnormalized positive map has output trace four. The illustration records this finite scaling distinction and does not assert a general recovery construction.